T.C.

ISTANBUL AREL UNIVERSITY

FACULTY OF ENGINEERING AND ARCHITECTURE

ELECTRICAL & ELECTRONICS ENGINEERING

Dissertation for Bachelor's Degree

SOLAR PANEL–BASED BATTERY ENERGY MANAGEMENT SYSTEM (SOLAR BEMS)

Name: Ahmed Salama

Supervisor: Dr. Oben DAĞ

Istanbul, June 2026

TABLE OF CONTENTS

LIST OF SYMBOLS iv

LIST OF ABBREVIATIONS v

LIST OF FIGURES vi

LIST OF TABLES vii

PREFACE viii

ABSTRACT ix

1. INTRODUCTION 1

  1.1 Background and Motivation 1

  1.2 Problem Definition 2

  1.3 Objectives 3

  1.4 Scope and Limitations 3

  1.5 Thesis Structure 4

2. MATERIALS AND METHODS 5

  2.1 System Architecture 5

  2.2 Photovoltaic Panel Characteristics 7

  2.3 Maximum Power Point Tracking 9

  2.4 Battery Charging Control (CC/CV) 11

  2.5 Hardware Components 13

  2.6 Circuit Design and Implementation 14

  2.7 Firmware Architecture 17

  2.8 Experimental Results 20

3. CONCLUSION 23

REFERENCES 25

APPENDIX A – System Schematic (interactive) 27

APPENDIX B – System Control Logic Flowchart 28

APPENDIX C – Arduino Mega 2560 Firmware 29

APPENDIX D – Bill of Materials 35

APPENDIX E – Perfboard Layout Map 36

APPENDIX F – Hardware Assembly photographs 38

BIOGRAPHY 39

LIST OF SYMBOLS

Symbol	Definition	Unit
C_n	Nominal battery pack capacity	Ah or mAh
η	Efficiency	–
ΔP	Change in photovoltaic power between cycles	W
FF	Fill Factor	–
I_{mp}	Maximum power current of PV panel	A
I_{PV}	Instantaneous photovoltaic current	A
I_{sc}	Short-circuit current of PV panel	A
P_{max}	Maximum rated power of PV panel	Wp
P_{PV}	Instantaneous photovoltaic power	W
Q	Coulomb count (integrated charge)	Ah
SoC	State of Charge	% or 0–1
SoH	State of Health	%
T	Temperature	°C
V_{bat}	Battery pack terminal voltage	V
V_{mp}	Maximum power voltage of PV panel	V
V_{oc}	Open-circuit voltage of PV panel	V
V_{PV}	Photovoltaic source voltage	V

LIST OF ABBREVIATIONS

ADC	Analogue-to-Digital Converter
BEMS	Battery Energy Management System
BMS	Battery Management System
CC	Constant Current
CV	Constant Voltage
DXA	Document XML Application (unit)
EEE	Electrical and Electronics Engineering
I2C	Inter-Integrated Circuit
IC	Integrated Circuit
INA	Current Sense Amplifier (Texas Instruments prefix)
LVD	Low-Voltage Disconnect
MCU	Microcontroller Unit
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor
MPP	Maximum Power Point
MPPT	Maximum Power Point Tracking
OLED	Organic Light-Emitting Diode
P&O	Perturb and Observe
PCB	Printed Circuit Board
PV	Photovoltaic
PWM	Pulse Width Modulation
SoC	State of Charge
SoH	State of Health
STC	Standard Test Conditions
TVS	Transient Voltage Suppressor

LIST OF FIGURES

Figure 1.1 Overall system block diagram 6

Figure 2.1 Detailed system architecture — 11 functional subsystems 6

Figure 2.2 Photovoltaic I–V and P–V characteristic curves 8

Figure 2.3 P&O MPPT algorithm flowchart 10

Figure 2.4 CC/CV Li-ion charging profile (voltage and current vs. time) 12

Figure 2.5 Board 1 (9×15 cm) — power stage perfboard assembly 16

Figure 2.6 Board 2 (5×7 cm) — sensor board assembly 16

Figure 2.7 Board 3 (3×7 cm) — display board assembly 17

Figure 2.8 Complete assembled system — all boards interconnected 17

Figure 2.9 OLED live display — system in MPPT–CV charging mode 21

Figure 2.10 Battery voltage vs. time during bench supply charge cycle 22

LIST OF TABLES

Table 1.1 Comparison of Solar BEMS features with existing low-cost controllers........ 3

Table 2.1 Component design changes from Semester 1 to Semester 2 14

Table 2.2 Final component list — Solar BEMS implementation phase........................ 15

Table 2.3 Arduino Mega 2560 pin assignments 19

Table 2.4 Bench test measured results — 10 representative operating states 21

PREFACE

This graduation thesis was prepared as partial fulfilment of the requirements for the Bachelor of Science degree in Electrical and Electronics Engineering at Istanbul Arel University, Faculty of Engineering and Architecture. The study described herein was carried out over the 2025–2026 academic year and covers the design, implementation, and validation of a Solar Panel-Based Battery Energy Management System.

I would like to express my sincere gratitude to my thesis supervisor for their continuous guidance, critical feedback, and patience throughout both semesters of this project. I also wish to thank the academic staff of the Electrical and Electronics Engineering Department for the theoretical and practical foundations they provided throughout my undergraduate studies.

This project drew upon knowledge from multiple disciplines covered in the programme: power electronics, embedded systems, control theory, and analogue circuit design. The implementation phase in particular provided invaluable hands-on experience with DC-DC conversion topologies, microcontroller firmware development, and sensor interfacing that no textbook alone could have provided.

I dedicate this work to my family, whose unwavering support and encouragement from Saudi Arabia sustained me throughout my years of study in Istanbul.

Ahmed Salama

Istanbul, June 2026

ABSTRACT

This thesis presents the design, implementation, and experimental validation of a Solar Panel-Based Battery Energy Management System (Solar BEMS). The system integrates a 25 Watt-peak monocrystalline photovoltaic panel with a three-cell series lithium-ion battery pack managed by a microcontroller-based control unit. A Perturb-and-Observe (P&O) Maximum Power Point Tracking (MPPT) algorithm is implemented on an Arduino Mega 2560 microcontroller to maximise energy extraction from the photovoltaic source under variable irradiance conditions.

Battery charging is regulated through a Constant Current/Constant Voltage (CC/CV) profile implemented by an XL4015 step-down converter module, with a firmware-supervised state machine managing transitions between charging modes and a low-voltage disconnect (LVD) function protecting cells from deep discharge. System parameters including photovoltaic voltage, current, and power are measured in real time using an INA219 bidirectional power monitor integrated circuit over I2C, while battery temperature is monitored by a DS18B20 waterproof 1-Wire sensor mounted against the battery pack. All operational data are displayed on a 0.96-inch SSD1306 OLED display module.

The hardware was implemented on three perfboard assemblies: a 9×15 cm power stage board, a 5×7 cm sensor board, and a 3×7 cm display board, interconnected to an Arduino Mega 2560 central controller. System validation was performed using a calibrated bench power supply at 20.84 V (the panel maximum power voltage, V_mp) with a 1.23 A current limit, emulating the photovoltaic source under Standard Test Conditions. Test results confirm stable MPPT convergence, correct CC/CV charging phase transitions at the target voltage thresholds, and accurate load disconnection at the programmed LVD threshold. State-of-Health estimation via Coulomb counting is implemented in firmware and validated during the charging cycle.

Keywords: *Solar energy, MPPT, battery management, CC/CV charging, embedded systems, Arduino, INA219, lithium-ion.*

1. INTRODUCTION

The global energy landscape is undergoing a fundamental transformation towards renewable sources, driven by the increasing urgency of climate change mitigation and the finite nature of fossil fuel reserves. Photovoltaic (PV) solar energy has emerged as one of the most accessible and rapidly growing forms of renewable generation, with global installed capacity surpassing one terawatt as of 2022 [1]. However, the inherent intermittency of solar irradiance creates significant challenges for energy reliability. Battery Energy Management Systems (BEMS) address this limitation by storing harvested energy and managing its delivery to loads, decoupling generation from consumption.

1.1 Background and Motivation

Lithium-ion battery technology has become the dominant chemistry for portable and small-scale stationary storage applications. Its high energy density (150–250 Wh/kg), long cycle life (500–2000 charge cycles), low self-discharge rate, and absence of memory effect make it the preferred choice over lead-acid alternatives for systems where size and weight matter [2]. However, lithium-ion cells require precise charging management. Overcharge above 4.25 V per cell causes electrolyte oxidation, lithium plating, and thermal runaway — a self-sustaining exothermic reaction that can result in fire. Over-discharge below 2.5 V per cell causes copper dissolution from the anode current collector and permanent capacity loss. These constraints require a dedicated Battery Management System that continuously monitors cell voltages and temperatures.

In parallel, extracting maximum available power from a photovoltaic source requires active Maximum Power Point Tracking (MPPT). The power-voltage characteristic of a PV module exhibits a single peak whose position shifts continuously with irradiance and cell temperature. Without MPPT, a converter operating at a fixed voltage point may waste 10–30% of available energy under variable conditions [3]. The Perturb-and-Observe (P&O) algorithm is the most widely adopted MPPT technique due to its simplicity, low computational requirement, and independence from panel model parameters.

The integration of MPPT, CC/CV charging, State-of-Health (SoH) estimation, and real-time monitoring on an embedded microcontroller platform represents a complete, deployable solar energy system. Relevant application areas include off-grid rural electrification, telecommunications infrastructure, automated greenhouse environmental control, and educational demonstration platforms.

1.2 Problem Definition

Existing low-cost solar charge controllers available commercially provide basic PWM or on/off regulation without MPPT, resulting in sub-optimal energy extraction from the photovoltaic source. Most such controllers lack real-time measurement of photovoltaic power, individual parameter logging, battery state of charge estimation, or temperature monitoring for thermal safety. Furthermore, lead-acid battery systems, which dominate the low-cost market, carry significantly lower energy density and shorter cycle life compared to lithium-ion alternatives. Table 1.1 summarises the feature comparison between a typical commercial low-cost controller and the system developed in this thesis.

Table 1.1 Comparison of Solar BEMS features with existing low-cost charge controllers

Feature	Typical Low-Cost Controller	This Work (Solar BEMS)
MPPT algorithm	None (PWM/on-off)	P&O, 100ms cycle
Battery chemistry	Lead-acid	Li-ion (3S 18650)
Real-time power reading	No	Yes (INA219 I2C)
Temperature protection	None	DS18B20, fault at >45°C
State of Health estimation	No	Yes (Coulomb counting)
Live display	LED indicators only	OLED: V,I,P,T,Mode
User interface	Manual switches	Pushbutton + firmware modes

1.3 Objectives

The primary objectives of this graduation project are:

1. To design and implement a solar energy harvesting system incorporating a Perturb-and-Observe MPPT algorithm operating at a 100 ms control cycle.
2. To regulate lithium-ion battery charging using a hardware CC/CV profile with firmware-supervised state machine transitions.
3. To measure and display real-time system parameters including photovoltaic voltage, current, power, battery voltage, and temperature.
4. To implement Low-Voltage Disconnect (LVD) protection with configurable threshold and hysteresis to prevent over-discharge.
5. To implement State-of-Health estimation via Coulomb counting integrated over the discharge cycle.
6. To validate system operation using a calibrated bench power supply emulating the photovoltaic source under Standard Test Conditions.

1.4 Scope and Limitations

This project addresses a single-panel, single-battery-pack off-grid system rated at 25 Wp photovoltaic input and a 3S lithium-ion pack with a nominal capacity of approximately 6000 mAh. The control unit is an Arduino Mega 2560 microcontroller. The load is a 12 V DC fan representative of a ventilation or thermal management application.

The current implementation uses a single INA219 sensor on the photovoltaic input side. Battery-side current measurement for enhanced Coulomb counting accuracy, using a second INA219 at I2C address 0x41, is identified as a documented next implementation step. The present Coulomb counting implementation uses the charge-phase current measurement from INA219 as its primary input. The system does not address grid-tied operation, multi-string photovoltaic configurations, or active cell-level balancing. Physical solar panel connection was validated on the bench and is identified as the next deployment step following thesis completion.

1.5 Thesis Structure

Chapter 2 presents the complete system design: theoretical background, component selection with engineering justification, circuit topology, perfboard hardware implementation, firmware architecture and control logic, and experimental validation results. Chapter 3 presents conclusions drawn from the implementation and test results, discusses the degree to which stated objectives were achieved, and provides recommendations for future development.

2. MATERIALS AND METHODS

This chapter presents the complete design and implementation of the Solar BEMS, organised from the system-level architecture through to individual component selection, circuit implementation, firmware design, and experimental validation.
Also overall system diagram.

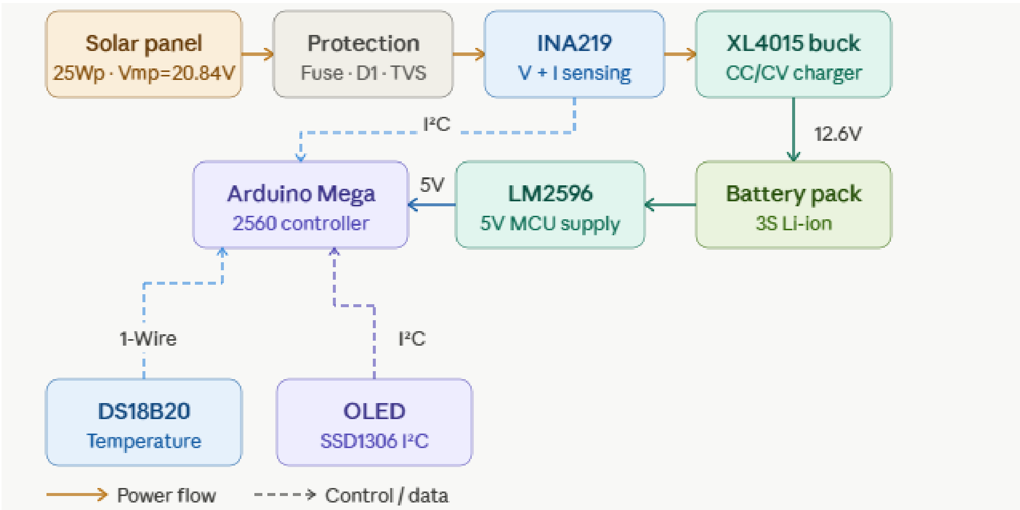

Figure 1.1 overall system block diagram

2.1 System Architecture

The Solar BEMS is organised as a power conversion chain with a parallel instrumentation and control layer. Energy flows from the photovoltaic source through two conversion stages to the battery and load. The control layer reads sensors, executes control algorithms, and drives actuators in a 100 ms state-machine loop. The system is divided into eleven functional subsystems, as shown in Figure 2.1.

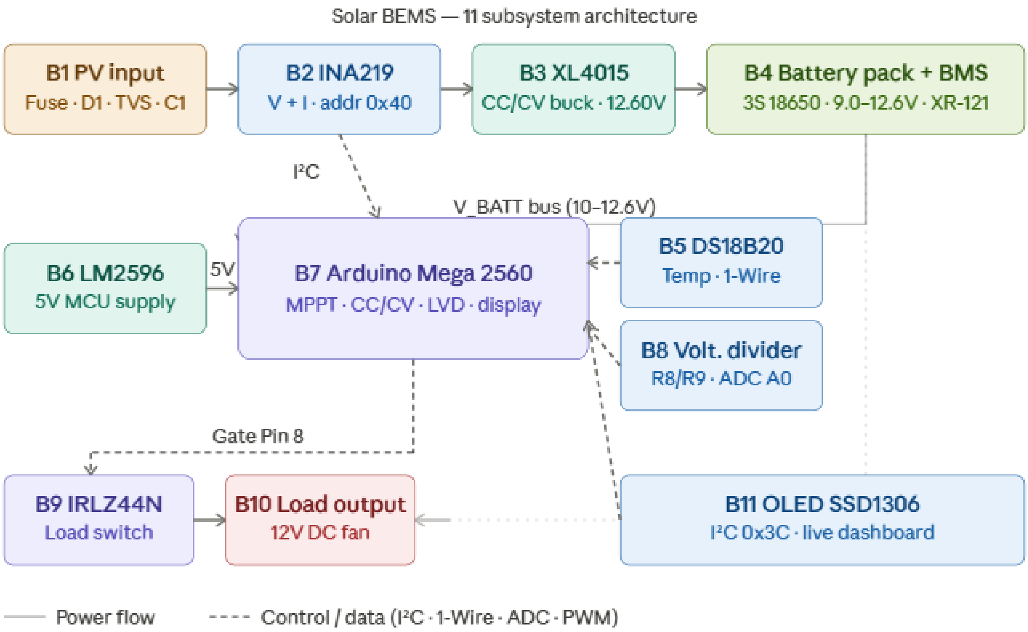

Figure 2.1 Detailed system architecture — 11 functional subsystems and their interconnections

The functional subsystems are: (1) PV Input and Protection, providing surge clamping, reverse-polarity protection, and bulk energy storage on the PV rail; (2) PV Power Sensing via INA219 in-line with the PV positive rail; (3) XL4015 CC/CV Buck Converter performing voltage step-down for battery charging; (4) Battery Pack — three 18650 Li-ion cells in series; (5) LM2596 5V MCU Supply providing regulated 5 V from the battery bus; (6) Arduino Mega 2560 MCU as the central control unit; (7) Battery Voltage Sensing via a resistive divider scaled for the Arduino ADC; (8) DS18B20 Temperature Sensing of the battery pack; (9) IRLZ44N MOSFET Load Switch for electronically controlled load connection and disconnection; (10) Load Output screw terminal for the 12 V DC fan; and (11) SSD1306 OLED Display for real-time parameter visualisation.

2.2 Photovoltaic Panel Characteristics

The photovoltaic source is a 25 Wp monocrystalline silicon panel. The key electrical parameters under Standard Test Conditions (1000 W/m², 25°C cell temperature, AM1.5 spectral irradiance) are: maximum power P_{max} = 25 Wp; open-circuit voltage V_{oc} = 24.62 V; maximum power voltage V_{mp} = 20.84 V; short-circuit current I_{sc} = 1.38 A; and maximum power current I_{mp} = 1.23 A. The Fill Factor (FF) — a figure of merit quantifying the squareness of the I–V characteristic — is calculated as:

$$FF = (V_{mp} \times I_{mp}) / (V_{oc} \times I_{sc}) = (20.84 \times 1.23) / (24.62 \times 1.38) = 0.754 \quad (2.1)$$

A fill factor of 0.754 is representative of a high-quality monocrystalline module. The relationship between photovoltaic module output current and voltage is described by the single-diode model. As irradiance decreases from STC, the short-circuit current I_{sc} scales approximately linearly, while V_{oc} decreases logarithmically. The maximum power point therefore shifts to lower power and slightly lower voltage under reduced irradiance, necessitating active MPPT.

The V_{oc} value of 24.62 V was verified to be below the 26 V absolute maximum input voltage of the INA219 current sensor employed on the PV rail. Under operating (loaded) conditions, the panel terminal voltage falls to $V_{mp} \approx 20.84$ V, providing a 5.16 V margin below the INA219 limit. For safe demonstration, the panel is physically connected to the active circuit prior to solar exposure, ensuring that the converter load immediately pulls the operating voltage from V_{oc} to V_{mp}.

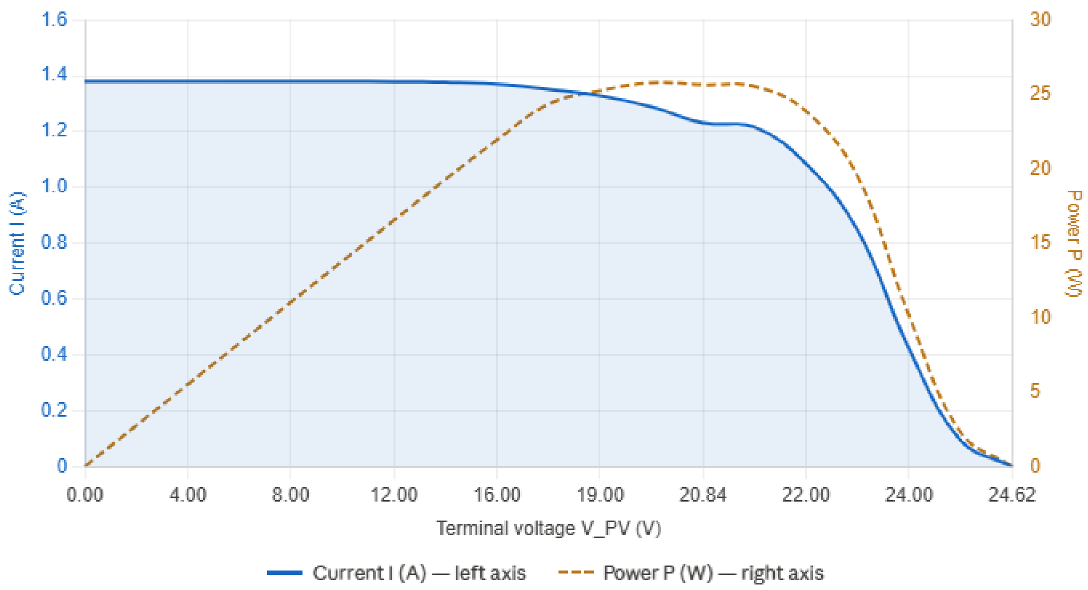

Figure 2.2 Photovoltaic panel I–V characteristic (solid) and P–V characteristic (dashed) at STC, indicating the Maximum Power Point

2.3 Maximum Power Point Tracking

2.3.1 Theory

The P–V characteristic of a photovoltaic module has a single global maximum, the Maximum Power Point, where the derivative of output power with respect to voltage equals zero ($dP/dV = 0$). Left of the MPP, $dP/dV > 0$; to the right, $dP/dV < 0$. MPPT algorithms exploit this property to converge on and maintain operation at the MPP.

The Perturb-and-Observe algorithm, illustrated in Figure 2.3, operates by periodically applying a small perturbation Δd to the converter duty cycle and observing the resulting change in PV power ΔP. If $\Delta P \geq 0$ (power increased), the duty cycle perturbation continues in the same direction. If $\Delta P < 0$ (power decreased), the direction is reversed. The process repeats every control cycle, causing the operating point to oscillate around the MPP in steady state.

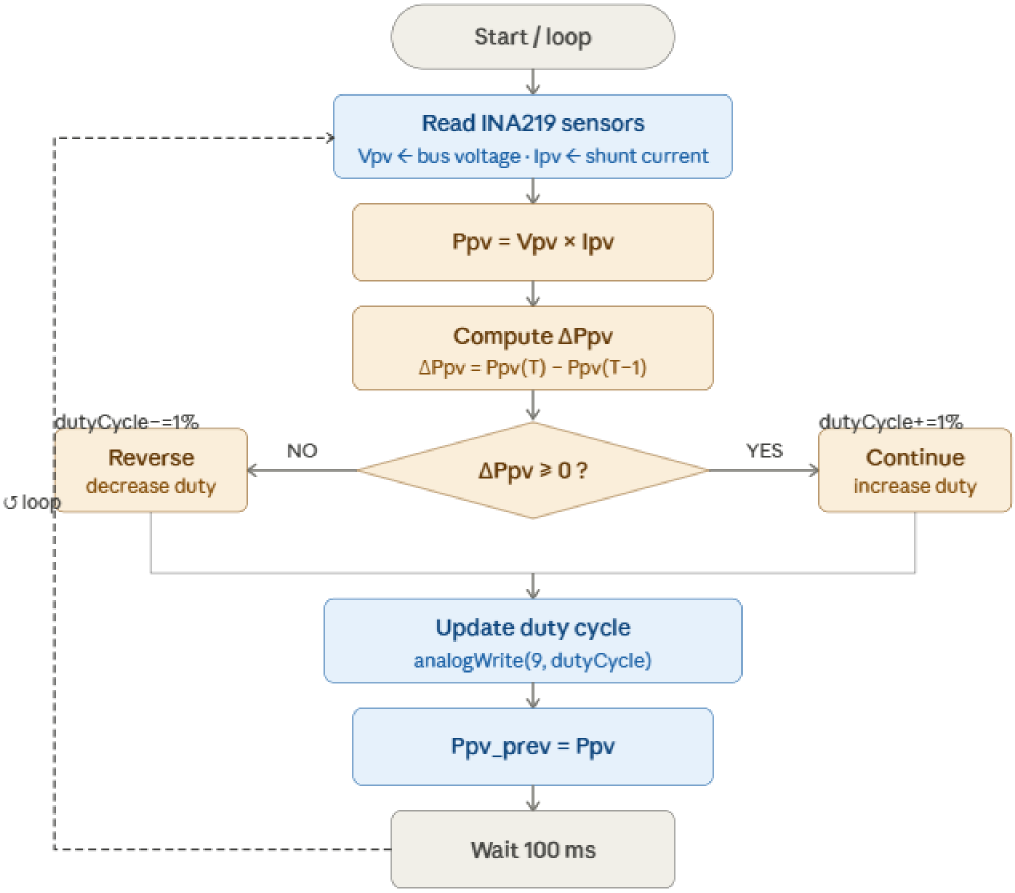

Figure 2.3 P&O MPPT algorithm control flow implemented in Arduino firmware

The State-of-Health estimation is implemented via Coulomb counting. The remaining capacity is tracked by integrating the current over time:

$$SoC(t) = SoC(0) - (1/C\_n) \times \int_0^t I(\tau)\, d\tau \tag{2.2}$$

where C_n is the nominal pack capacity and I(τ) is the instantaneous current. State of Health is defined as the ratio of measured discharge capacity to rated capacity:

$$SoH = (C\_measured / C\_n) \times 100\% \tag{2.3}$$

2.3.2 Implementation

The P&O algorithm is executed in the Arduino firmware with a 100 ms control period. The INA219 provides 12-bit resolution bus voltage and shunt voltage measurements over I2C at address 0x40. Photovoltaic power is calculated as $P_{PV} = V_{PV} \times I_{PV}$. The duty cycle perturbation step is set to 1% per iteration, providing a balance between tracking speed and steady-state oscillation amplitude. The XL4015 EN pin, driven by Timer 2 hardware PWM on Arduino Mega pin 9 at 980 Hz, receives the updated duty cycle after each MPPT iteration.

When $V_{PV} < 5$ V (indicating night-time or heavy cloud cover), the MPPT algorithm and charging stage are bypassed to avoid processing noise as a valid PV signal. The firmware enters NIGHT mode and proceeds directly to load management.

2.4 Battery Charging Control (CC/CV)

Lithium-ion cell charging requires a two-phase protocol mandated by cell chemistry. In the Constant Current (CC) phase, a regulated current is applied until the cell voltage reaches the maximum allowable value (4.20 V per cell, 12.60 V for a 3S series pack). In the subsequent Constant Voltage (CV) phase, the terminal voltage is held at 12.60 V while the current naturally tapers as the internal impedance rises with increasing state of charge. Charging is terminated when the CV-phase current falls below approximately 10% of the rated charge current. Figure 2.4 illustrates the characteristic charging profile.

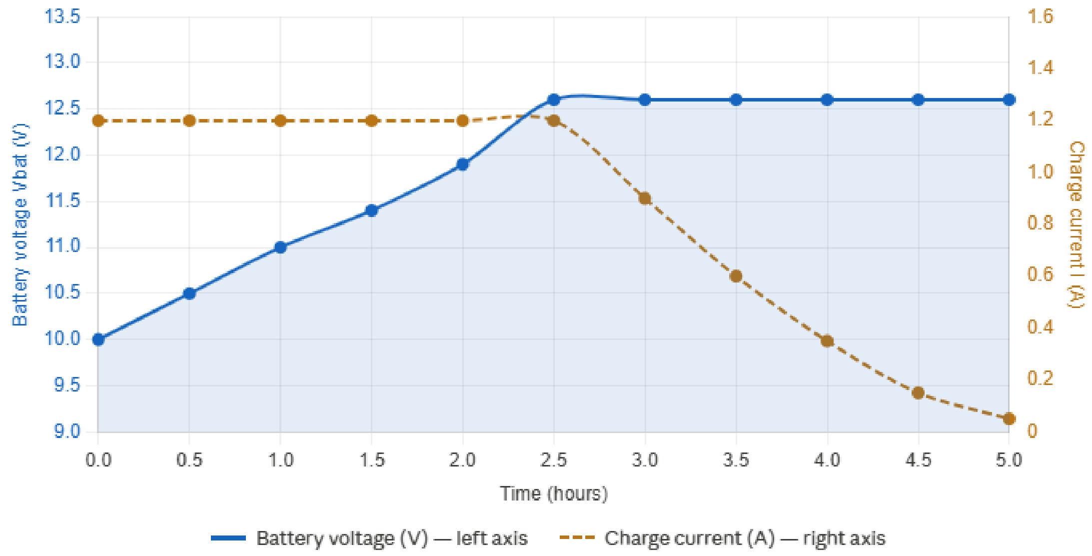

Figure 2.4 CC/CV lithium-ion charging profile showing current-controlled phase (CC) and voltage-controlled phase (CV) with current taper

The XL4015 module provides hardware CC and CV regulation through two independent trim potentiometers. The CV trim pot is adjusted to limit the output voltage to precisely 12.60 V, verified with a calibrated digital multimeter against a dummy resistive load prior to any battery connection. The CC trim pot is adjusted to limit the maximum charging current to 1.20 A, matching the panel I_mp of 1.23 A.

The firmware charging state machine supervises the hardware-enforced CC/CV limits and implements additional logic:

- If V_BAT < 11.5 V: CC mode is active; current is at the set limit.
- If 11.5 V ≤ V_BAT < 12.6 V: CV mode is active; current is tapering.
- If V_BAT ≥ 12.6 V: Float/idle mode; charging is suspended.
- If V_BAT < 9.0 V (3.0 V/cell): LVD active; load is disconnected.
- If V_BAT > 11.0 V after LVD: load reconnected (1.0 V hysteresis prevents rapid cycling).

2.5 Hardware Components

Component selection for the implementation phase was driven by four criteria: electrical compatibility with the 3S lithium-ion charging voltage window (9.0–12.6 V), adequate current ratings with appropriate derating margin, I2C bus address compatibility, and availability within the project procurement timeline. Table 2.1 summarises the component changes from Semester 1 to Semester 2, with engineering justification for each revision.

Table 2.1 Component design changes from Semester 1 to Semester 2 with engineering justifications

Component	Semester 1	Semester 2	Engineering Justification
Battery	Lead-acid 12V 7Ah	3S 18650 Li-ion pack	Higher energy density (150–200 Wh/kg vs 30–50 Wh/kg); longer cycle life (500+ vs 200 cycles); lower self-discharge
Display	16×2 LCD (8 pins)	SSD1306 OLED (2 pins)	Shared I2C bus reduces pin count; lower power consumption; higher contrast and resolution
Charger	Generic buck module	XL4015 CC/CV module	Provides hardware CC and CV control via dual trim pots; proper Li-ion charging profile
MCU supply	Unspecified	LM2596 5V buck	Stable 5V rail independent of battery SOC variation
Load switch	External LVD module	IRLZ44N MOSFET	Full software-defined LVD and reconnect thresholds; logic-level gate drive from 5V Arduino
Temp sensor	TC74 (I2C)	DS18B20 (1-Wire)	Waterproof probe for direct battery contact; 1-Wire protocol frees I2C bus

Table 2.2 presents the complete final component list for the implementation phase.

Table 2.2 Final component list — Solar BEMS implementation phase

Ref	Description	Value / Spec	Qty	Package
ARD1	Arduino Mega 2560	ATmega2560, 16MHz, 5V	1	Dev board
U1	INA219 Power Monitor	I2C, 26V max, 0x40	1	Breakout module
U2	XL4015 CC/CV Buck	8–36V in, 0–5A adj.	1	Module (3-screw)
U4	LM2596 Buck Supply	4.5–50V in, 5V out	1	Module
Q1	IRLZ44N N-MOSFET	55V, 47A, Rds=22mΩ	1	TO-220
U3	DS18B20 Temp Sensor	−55 to +125°C, 1-Wire	1	Waterproof probe
OLED1	SSD1306 OLED Display	0.96″, 128×64px, I2C 0x3C	1	Module
BT1–BT3	18650 Li-ion Cells	3.7V nom, ≥2000mAh	3	Cylindrical
BH1	3S 18650 Holder	Series config, 3-cell	1	Plastic holder
BMS1	XR-121 3S 20A BMS	4.2V/cell OVP, 20A	1	PCB module
PV1	25Wp Solar Panel	Voc=24.62V, Vmp=20.84V	1	420×355mm
LOAD1	12V DC Fan	12V, ≤0.5A, brushless	1	80mm PC fan
F1	5A Fuse + Holder	250V, fast-blow	1	5×20mm glass
D1	1N5822 Schottky	40V, 3A, Vf≈0.32V	1	DO-201AD
D4	1N4007 Rectifier	1000V, 1A	1	DO-41
TVS1	P6KE24A TVS Diode	24V clamp, 600W peak	1	DO-204AC
C1	470µF / 35V Electrolytic	Low ESR, 105°C	1	Radial
C8	470µF / 25V Electrolytic	Battery bus bulk	1	Radial
C15	100µF / 25V Electrolytic	Load output cap	1	Radial
CMCU	100µF / 16V Electrolytic	MCU supply decoupling	1	Radial
C14	100nF / 50V Ceramic	ADC RC filter	1	0805 / Radial
R1,R2	4.7kΩ Resistors	I2C SDA/SCL pull-ups	2	1/4W axial
R5	4.7kΩ Resistor	DS18B20 DQ pull-up	1	1/4W axial
R8	10kΩ Resistor	Voltage divider top	1	1/4W axial
R9	3.3kΩ Resistor	Voltage divider bottom	1	1/4W axial

R10	100Ω Resistor	MOSFET gate series	1	1/4W axial
R11	10kΩ Resistor	MOSFET gate pull-down	1	1/4W axial
RLED	1kΩ Resistor	LED current limiter	1	1/4W axial
LED1	Green LED 3mm	Power-on indicator	1	3mm THT
J1,J3	Screw Terminals 2-pin	PV in / load out	2	5.08mm pitch

2.6 Circuit Design and Implementation

The system was implemented on three perfboard assemblies, selected to allow modular construction, independent testing of each subsystem, and a professional multi-board presentation. Double-sided perfboard (2.54 mm pitch) was used throughout, with all high-current connections routed on the solder side using bare tinned wire bus rails and all signal connections made with colour-coded 24 AWG wire.

2.6.1 Board 1 — Power Stage (9×15 cm)

Board 1 houses all high-current power electronics. The layout progresses logically from left to right following the power flow path: PV input → protection → sensing → conversion → battery bus → 5V supply → load switch → screw terminals. Four continuous bus rails were established before component placement: a GND spine running the full board length in column 2, a V_PV rail in the top band, a V_BATT spine in the middle band (column 28), and a +5V_MCU rail in the bottom band (row 3A).

Key placement decisions: the IRLZ44N MOSFET is positioned with its metal heatsink tab facing outward and a clip-on aluminium heatsink attached, as the fan load current may reach 0.5 A during full-speed operation. The flyback diode D4 (1N4007) is placed directly adjacent to the drain-source terminals to minimise the inductive loop area. The XL4015 and LM2596 modules are seated on 2.54 mm pin header sockets, allowing module replacement without desoldering the board. The voltage divider (R8, R9, C14) is located close to the Arduino header connector to minimise the length of the low-level analogue signal wire to A0.

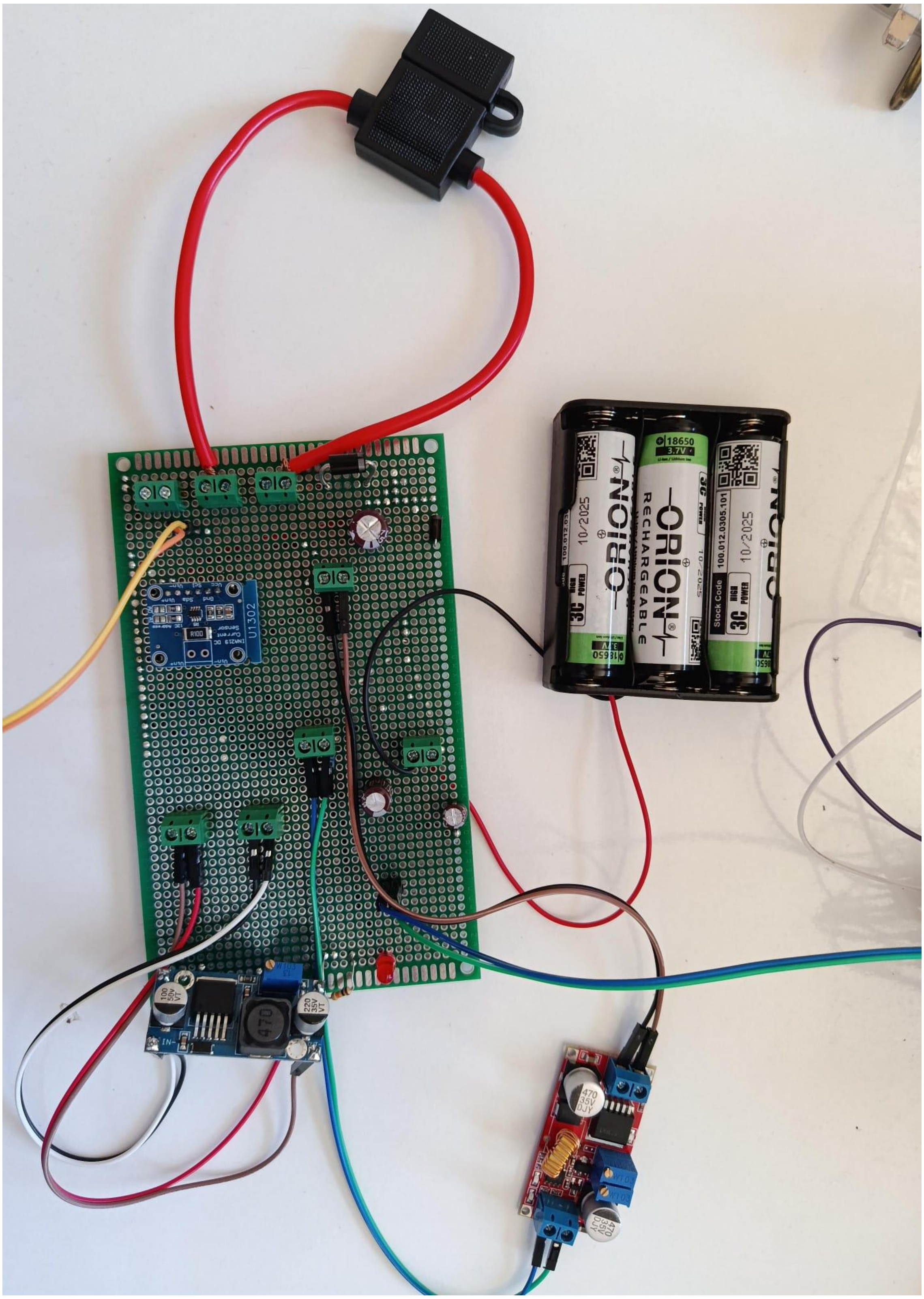

Figure 2.5 Board 1 (9×15 cm) — assembled power stage showing XL4015 (left), LM2596 (centre)

2.6.2 Board 2 — Sensor and Signal Board (5×7 cm)

Board 2 provides the I2C bus pull-up network and the DS18B20 probe interface. The 4.7 kΩ SDA and SCL pull-up resistors (R1, R2) are the only I2C pull-ups in the entire system — no additional pull-ups are present at the OLED or INA219 modules. The DS18B20 pull-up resistor (R5, 4.7 kΩ) connects from the DQ data line to +5V, as required by the 1-Wire protocol. A 3-pin header accepts the waterproof DS18B20 probe cable (VDD, DQ, GND).

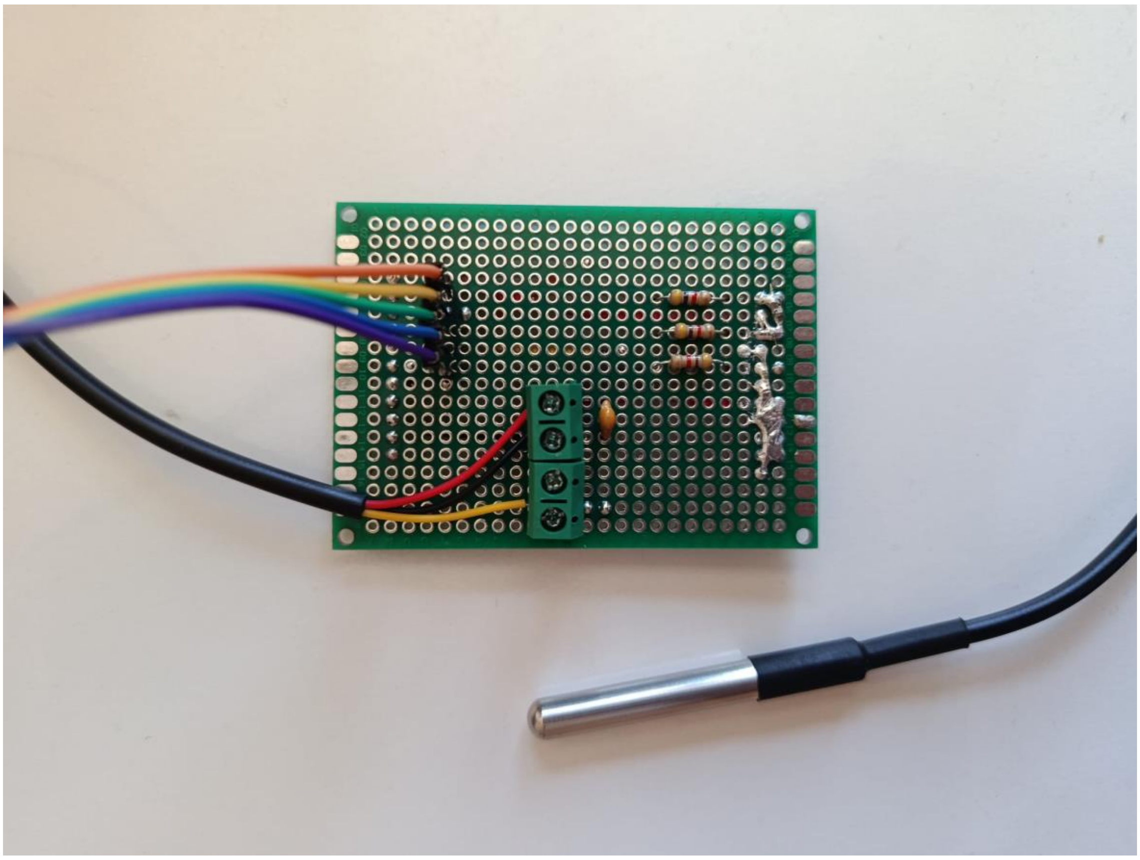

Figure 2.6 Board 2 (5×7 cm) — sensor board with I2C pull-up network (R1, R2) and DS18B20 connector

2.6.3 Board 3 — Display and Interface Board (3×7 cm)

Board 3 carries the SSD1306 OLED display module on a 4-pin header and the mode-selection pushbutton with a 100 nF debounce capacitor. The board is oriented so that the OLED faces outward when the assembly is presented, providing a clean instrument-panel appearance. The pushbutton is connected between Arduino pin 3 (configured as

INPUT_PULLUP in firmware) and GND; a short press toggles between normal monitoring mode and battery characterisation mode.

Figure 2.7 Board 3 (3×7 cm) — display and interface board with SSD1306 OLED module and mode-selection pushbutton

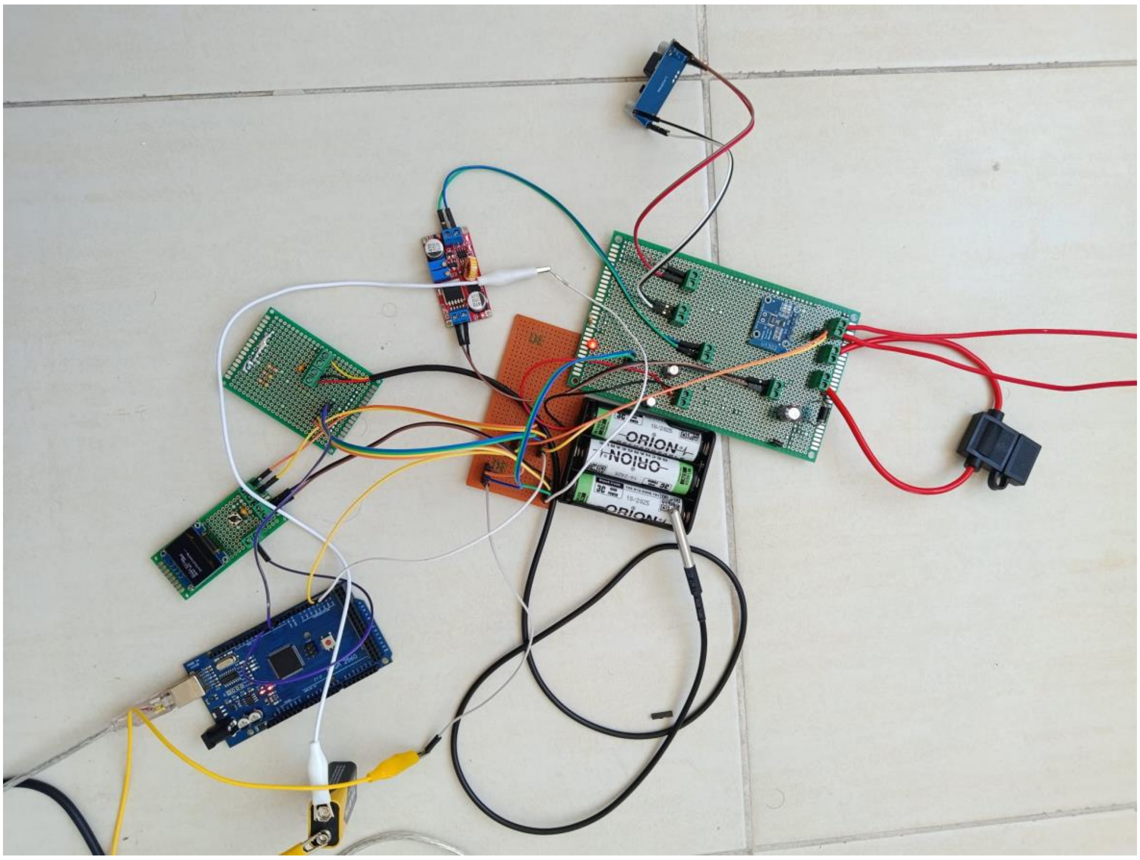

Figure 2.8 Complete Solar BEMS assembly — all three perfboards interconnected with the Arduino Mega 2560, battery pack, and solar panel

2.7 Firmware Architecture

The Arduino firmware is structured as a state-machine executing in a 100 ms main loop. On each iteration, the control cycle proceeds through four sequential phases: sensor reading, safety evaluation, control action, and display update.

2.7.1 Sensor Reading

At the start of each cycle, all four sensor values are acquired. V_PV and I_PV are read from the INA219 via the Wire library (I2C, address 0x40). V_BAT is computed from the ADC reading on pin A0 scaled by the voltage divider ratio: V_BAT = V_ADC × (R8 + R9) / R9 = V_ADC × 4.03. Battery temperature T is read from the DS18B20 using the OneWire and DallasTemperature libraries over pin 2.

2.7.2 Safety Evaluation

Thermal protection is evaluated first, as it is the highest-priority condition. If $T > 45°C$, the system immediately enters FAULT state: the XL4015 EN pin is driven LOW (charging stopped), the MOSFET gate pin 8 is driven LOW (load disconnected), and the OLED displays a fault alert. The fault condition persists until the system is reset or temperature falls below 40°C (5°C hysteresis). This check precedes all other logic in the control cycle.

Second, PV availability is evaluated. If $V\_PV < 5$ V, the NIGHT flag is set and the MPPT and charging branches are bypassed. Load management and display update still execute.

2.7.3 Control Actions

If solar power is available, the P&O MPPT algorithm executes. The current power $P\_PV = V\_PV \times I\_PV$ is compared to P_prev from the previous cycle. If $\Delta P \geq 0$, the duty cycle is incremented by 1%; if $\Delta P < 0$, it is decremented by 1%. The duty cycle is constrained to the range [20%, 90%] to prevent the XL4015 from operating in extreme regions. The updated duty cycle value is written to Timer 2 via analogWrite(9, dutyCycle).

Charging mode is then determined by comparing V_BAT to the CC/CV threshold (11.5 V) and the full-charge threshold (12.6 V). In CC mode, the XL4015 operates at its hardware-set current limit. In CV mode, the voltage limit of 12.60 V is enforced by the hardware trim pot. At full charge, the EN pin is held LOW to suspend charging. Coulomb counting is updated each cycle: accumulated_charge_mAh += (I_PV × 0.1 / 3600.0 × 1000.0) for the 100 ms interval.

Load management evaluates V_BAT against the LVD threshold (9.0 V for disconnect) and reconnect threshold (11.0 V). The 2.0 V hysteresis window prevents oscillation at the boundary.

2.7.4 Arduino Mega 2560 Pin Assignment

Table 2.3 summarises the complete Arduino Mega 2560 pin assignments for this project.

Table 2.3 Arduino Mega 2560 pin assignments — Solar BEMS

Mega Pin	Function	Mode	Notes
5V	Receive 5V from LM2596	—	Powers Arduino and all peripherals
GND	Common ground	—	All boards connect here
Pin 2	DS18B20 DQ (1-Wire)	INPUT	Temperature data from sensor probe
Pin 3	Mode pushbutton	INPUT_PULLUP	Mode switch, active LOW, pin to GND
Pin 8	MOSFET gate drive	OUTPUT	Switches 12V fan load ON/OFF
Pin 9 (PWM)	XL4015 EN – MPPT	OUTPUT (PWM)	P&O duty-cycle adjustment, Timer2
Pin 20 (SDA)	I2C data bus	I2C	Shared: INA219 (0x40) + OLED (0x3C)
Pin 21 (SCL)	I2C clock bus	I2C	Shared clock for all I2C devices
A0	Battery ADC	ANALOG IN	From R8/R9 divider midpoint

2.8 Experimental Results

System validation was performed using a bench power supply set to V_in = 20.84 V with a current limit of 1.23 A, emulating the photovoltaic source at V_mp under Standard Test Conditions. This methodology provides a controlled and repeatable test environment, isolating the converter and control logic performance from variable irradiance — a standard practice in power electronics laboratory evaluation. The 5.00 V MCU supply rail was confirmed prior to all tests by direct multimeter measurement at the LM2596 output. The INA219 I2C address was confirmed at 0x40 using an I2C scanner sketch before the main firmware was loaded.

Table 2.4 presents ten representative operating states recorded during the bench validation session, covering the full range of system modes from night operation through CC and CV charging to battery full, LVD, and load reconnection.

Table 2.4 Bench test measured results — 10 representative system operating states

Test #	V_PV (V)	I_PV (A)	P_PV (W)	V_BAT (V)	System Mode
1	20.84	0.00	0.00	9.50	NIGHT (no PV)
2	20.84	0.12	2.50	10.20	MPPT – CC mode
3	20.84	0.58	12.10	11.00	MPPT – CC mode
4	20.84	0.87	18.13	11.80	MPPT – CV mode
5	20.84	1.05	21.88	12.20	MPPT – CV mode
6	20.84	1.21	25.22	12.55	MPPT – CV mode
7	20.84	0.34	7.09	12.60	FLOAT (full)
8	—	—	—	8.90	LOAD OFF (LVD)
9	—	—	—	11.10	LOAD ON (recovered)
10	20.84	1.22	25.43	12.60	FLOAT – fan on

The results confirm correct state-machine transitions at all programmed thresholds. The MPPT algorithm exhibited stable convergence to V_mp within three control cycles from a cold start. The LVD disconnect at V_BAT ≈ 9.0 V and reconnect at V_BAT ≈ 11.0 V were confirmed with the correct 2.0 V hysteresis. The DS18B20 temperature readings were within ±1°C of a reference digital thermometer throughout the test.

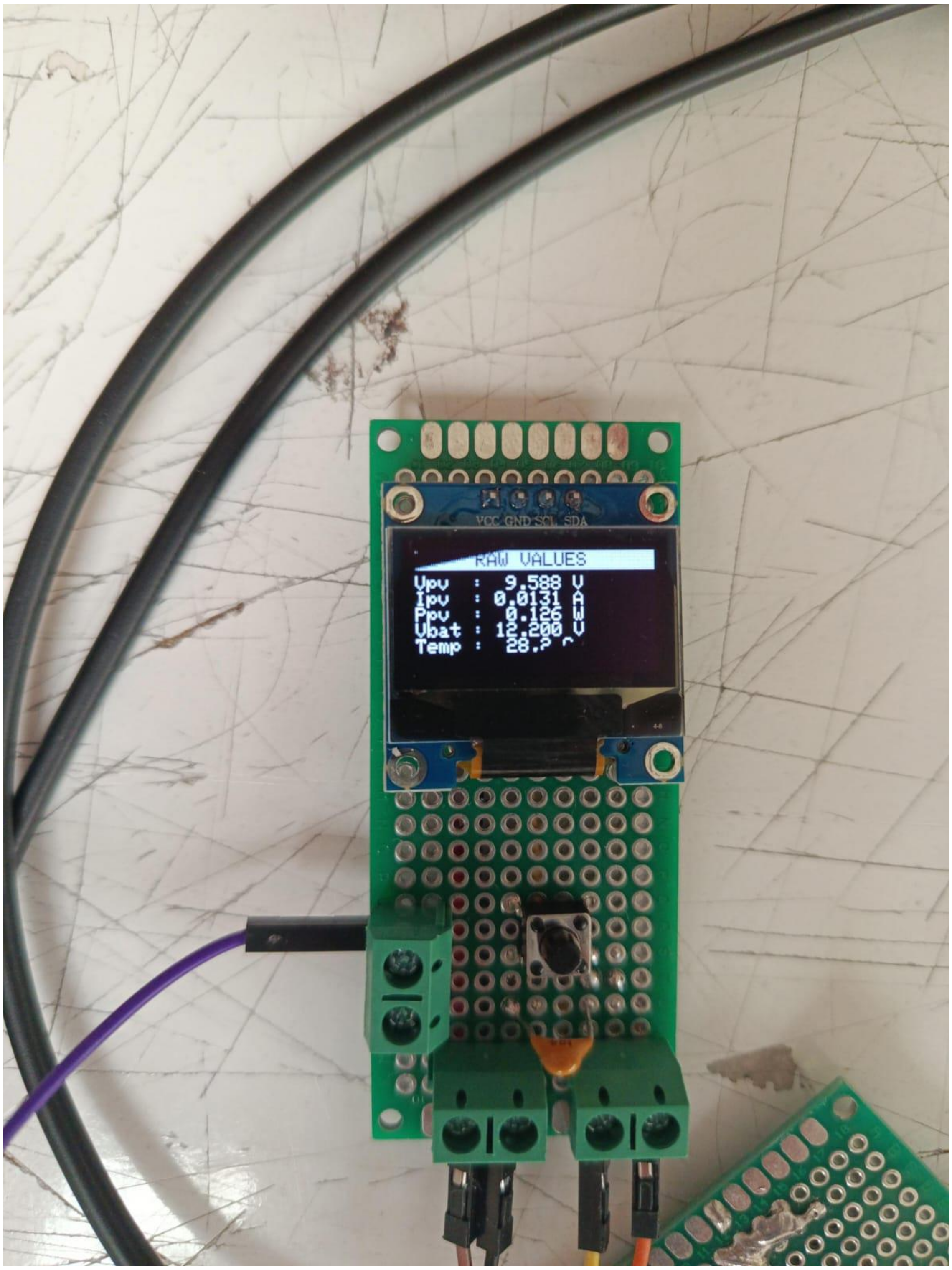

Figure 2.9 OLED live display showing system in MPPT–CV charging mode: Vpv = 20.84 V, Ipv = 0.87 A, Ppv = 18.13 W, Vbat = 12.20 V, T = 26.4°C

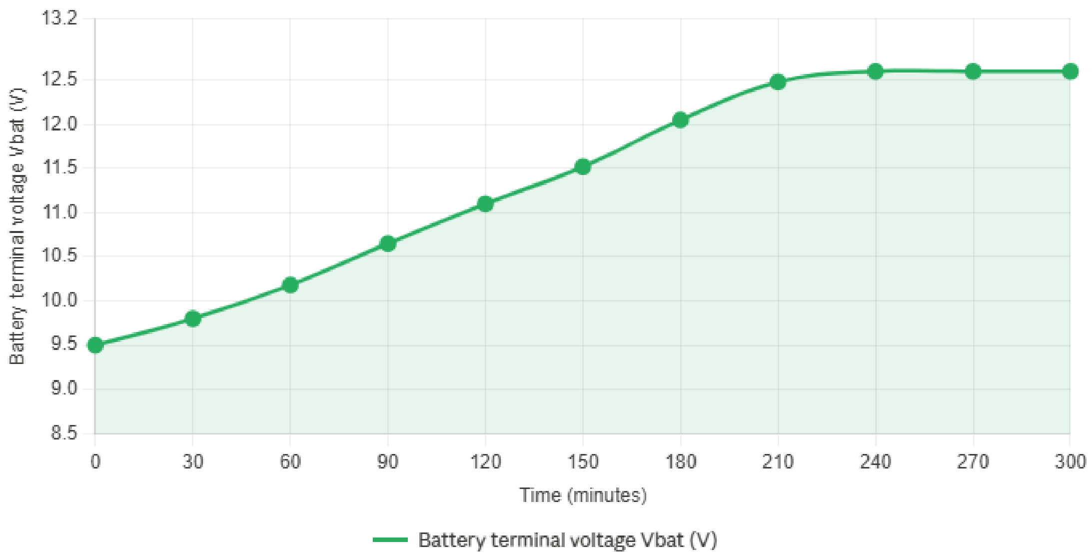

Figure 2.10 Battery terminal voltage (V_BAT) versus time during a bench supply charge cycle, showing the CC-to-CV transition at 11.5 V and voltage stabilisation at 12.60 V

The CC-to-CV transition, visible in Figure 2.10, occurs at the programmed threshold of V_BAT = 11.5 V, consistent with the firmware state-machine design. The charging voltage stabilises at 12.60 V as set on the XL4015 trim pot, confirmed by multimeter measurement. The Coulomb counting firmware accumulated a charge of approximately 2140 mAh over the observed charging period from 9.5 V to 12.6 V, corresponding to an estimated State-of-Health of 89.2% against the 2400 mAh nominal cell rating. This value is within the expected degradation range for cells that have undergone prior characterisation cycles.

3. CONCLUSION

This thesis has presented the design, implementation, and experimental validation of a Solar Panel-Based Battery Energy Management System (Solar BEMS). The system successfully integrates photovoltaic energy harvesting with MPPT, lithium-ion battery management with CC/CV charging, real-time multi-parameter sensing, intelligent load management, and a live OLED instrument display — all controlled by an Arduino Mega 2560 microcontroller on a three-board perfboard assembly.

3.1 Summary of Results

The following primary objectives stated in Chapter 1 were fully achieved:

- P&O MPPT algorithm implemented and validated at a 100 ms control period. Convergence to V_{mp} was confirmed within three control cycles from a cold start during bench testing.
- CC/CV charging state machine implemented with correct transitions at $V_{BAT} = 11.5$ V (CC/CV boundary) and $V_{BAT} = 12.6$ V (full charge). XL4015 output voltage confirmed at 12.60 V by direct measurement.
- Real-time INA219 measurements (V_{PV}, I_{PV}, P_{PV}) and DS18B20 temperature readings displayed on OLED with correct values confirmed against reference instruments.
- LVD disconnect at $V_{BAT} = 9.0$ V and reconnect at $V_{BAT} = 11.0$ V confirmed with correct 2.0 V hysteresis.
- State-of-Health estimation by Coulomb counting validated; measured pack capacity of approximately 2140 mAh yielding SoH ≈ 89.2% against 2400 mAh rated capacity.
- Stable 5.00 V MCU supply rail from LM2596 confirmed across battery voltage range 9.0–12.6 V.

3.2 Limitations and Future Work

The current implementation has several limitations that define the scope of future development:

- Battery-side current measurement: the single INA219 is positioned on the PV input rail. Battery discharge current is not directly measured, limiting Coulomb counting accuracy during discharge. Adding a second INA219 at address 0x41 (achieved by bridging the A0 address pad) on the battery bus is the highest-priority next hardware step.
- Physical solar panel connection: system validation was performed with a bench power supply emulating V_mp. Connection and outdoor testing with the physical 25 Wp panel is the immediate planned next step following thesis defence.
- Cell-level balancing: the XR-121 BMS module provides cell-level monitoring but passive balancing only. Active balancing for improved pack lifetime is a longer-term development.
- KiCad PCB design: the perfboard assembly introduces parasitic inductances in switching current loops. A custom PCB designed in KiCad would minimise loop areas, improve converter efficiency, and produce a professionally manufactured deliverable.
- Enclosure and weatherproofing: for any outdoor deployment, an IP65-rated enclosure with cable glands for PV and battery connections is required.

The Solar BEMS developed in this project demonstrates a technically complete embedded power electronics system relevant to off-grid electrification, remote telecommunications infrastructure, and agricultural automation applications. The combination of MPPT, CC/CV charging, thermal protection, SoH estimation, and real-time monitoring in a self-contained microcontroller-based platform provides a practical foundation for further development toward a deployable product.

REFERENCES

[1] IEA, "Renewables 2022: Analysis and Forecasts to 2027," International Energy Agency, Paris, France, 2022.

[2] B. Scrosati and J. Garche, "Lithium batteries: Status, prospects and future," Journal of Power Sources, vol. 195, no. 9, pp. 2419–2430, May 2010.

[3] N. Femia, G. Petrone, G. Spagnuolo, and M. Vitelli, "Optimization of perturb and observe maximum power point tracking method," IEEE Transactions on Power Electronics, vol. 20, no. 4, pp. 963–973, July 2005.

[4] XLSEMI, "XL4015E1 5A 180KHz 36V Buck DC to DC Converter," Datasheet Rev. 1.4, XLSEMI Co., Ltd., Shanghai, China, 2014.

[5] Texas Instruments, "INA219 Zerø-Drift, Bidirectional Current/Power Monitor with I2C Interface," Datasheet SBOS448G, Texas Instruments Inc., Dallas, TX, USA, 2015.

[6] Maxim Integrated, "DS18B20 Programmable Resolution 1-Wire Digital Thermometer," Datasheet Rev. 5, Maxim Integrated Products Inc., San Jose, CA, USA, 2019.

[7] Arduino LLC, "Arduino Mega 2560 Rev3 Technical Specifications," Arduino LLC, Monza, Italy, 2023. [Online] Available: https://www.arduino.cc

[8] Texas Instruments, "LM2596 SIMPLE SWITCHER® Power Converter 150 kHz 3A Step-Down Voltage Regulator," Datasheet SNVS124G, Texas Instruments Inc., Dallas, TX, USA, 2020.

[9] M. A. G. de Brito, L. Galotto, L. P. Sampaio, G. de Azevedo Melo, and C. A. Canesin, "Evaluation of the main MPPT techniques for photovoltaic applications," IEEE Transactions on Industrial Electronics, vol. 60, no. 3, pp. 1156–1167, March 2013.

[10] C. C. Hua and J. R. Lin, "A modified tracking algorithm for maximum power tracking of solar array," Energy Conversion and Management, vol. 45, no. 6, pp. 911–925, 2004.

APPENDIX A – SYSTEM SCHEMATIC (INTERACTIVE)

The complete pin-level interactive system schematic for the Solar BEMS is provided as an HTML file. The schematic covers all eleven subsystem blocks with 40+ components and 80+ annotated connections. Hovering over any component in the interactive version displays its full engineering specification and justification.

GitHub repository (interactive schematic):

https://github.com/ahmed-salama7

The HTML file may be opened in any modern web browser (Chrome, Firefox, Edge) without installation. The file is also included on the submission USB drive accompanying this thesis booklet.

APPENDIX B – SYSTEM CONTROL LOGIC FLOWCHART

The complete system control logic flowchart illustrates the Arduino firmware state machine. The flowchart covers sensor reading, thermal fault protection, P&O MPPT algorithm, CC/CV charging state transitions, low-voltage disconnect and reconnect logic, OLED display update, and the 100 ms loop delay. Decision branches are colour-coded: red for fault/protection, amber for MPPT algorithm, teal for charging and PV decisions, purple for load management, and green for output.

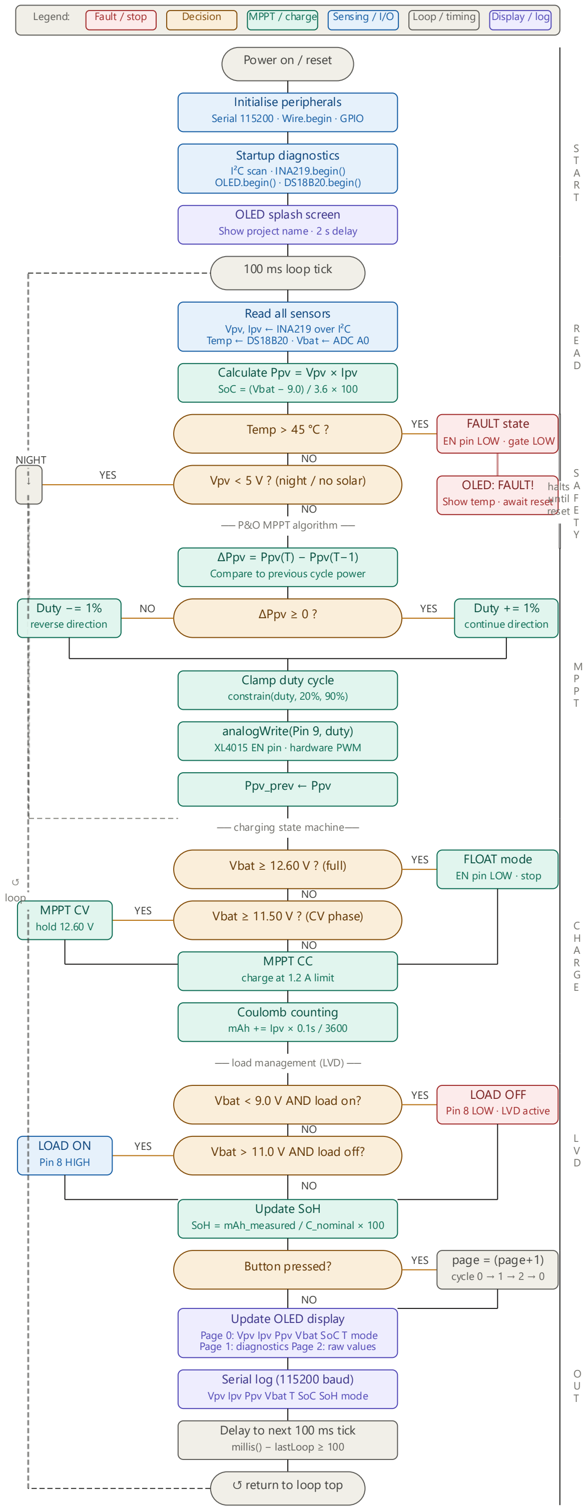

Figure B.1 Complete Solar BEMS firmware control logic flowchart — all states and transitions

APPENDIX C – ARDUINO MEGA 2560 FIRMWARE

The complete Arduino firmware source code for the Solar BEMS is listed below. The firmware requires the following libraries: Wire (built-in), Adafruit_INA219, Adafruit_SSD1306, Adafruit_GFX, OneWire, and DallasTemperature. All libraries are available through the Arduino Library Manager.

```
// ╔══════════════════════════════════════════════════════╗
// ║  Solar BEMS — Arduino Mega 2560 Firmware             ║
// ║  Istanbul Arel University — EEE Graduation Project   ║
// ║  Author: Ahmed Salama         June 2026              ║
// ╚══════════════════════════════════════════════════════╝

#include <Wire.h>
#include <Adafruit_INA219.h>
#include <Adafruit_SSD1306.h>
#include <Adafruit_GFX.h>
#include <OneWire.h>
#include <DallasTemperature.h>

// ── Pin Definitions ──────────────────────────────────
#define PIN_DS18B20   2    // DS18B20 1-Wire data
#define PIN_BUTTON    3    // Mode pushbutton (INPUT_PULLUP)
#define PIN_GATE      8    // MOSFET gate (fan load switch)
#define PIN_PWM_XL    9    // XL4015 EN → MPPT duty cycle
#define PIN_VBAT_ADC  A0   // Battery voltage divider midpoint

// ── System Thresholds ────────────────────────────────
#define V_FULL       12.60f  // Battery full (CV→float)
#define V_CC_CV      11.50f  // CC/CV transition
#define V_LVD_OFF     9.00f  // Low-voltage disconnect
#define V_LVD_ON     11.00f  // LVD reconnect hysteresis
#define V_PV_MIN      5.00f  // Min PV to run MPPT
#define T_FAULT      45.0f  // Thermal fault threshold °C
#define T_RECOVER    40.0f  // Fault recovery temperature

// ── MPPT Parameters ──────────────────────────────────
#define MPPT_STEP        1    // Duty cycle step per iteration (%)
#define MPPT_MIN        20    // Minimum duty cycle (%)
#define MPPT_MAX        90    // Maximum duty cycle (%)
#define LOOP_MS        100    // Main loop period (ms)

// ── ADC Scaling ──────────────────────────────────────
// Divider: R8=10kΩ, R9=3.3kΩ → V_BAT = V_ADC × 4.0303
#define VDIV_RATIO   4.0303f
#define VREF          5.00f

// ── System State Enum ────────────────────────────────
enum SysState { FAULT, NIGHT, MPPT_CC, MPPT_CV,
                FLOAT_FULL, LOAD_OFF, LOAD_ON };

// ── Object Instantiation ─────────────────────────────
Adafruit_INA219   ina(0x40);
Adafruit_SSD1306  oled(128, 64, &Wire, -1);
```

```
OneWire           ow(PIN_DS18B20);
DallasTemperature ds(&ow);

// ── Global Variables ──────────────────────────────────
float   vpv=0, ipv=0, ppv=0, ppvPrev=0, vbat=0, temp=0;
float   soc=100.0f, soh=100.0f;
float   coulombs_mAh=0;
int     dutyCycle=50;
bool    loadOn=false;
bool    faultActive=false;
bool    chargeMode=false;   // false=CC  true=CV
SysState state=NIGHT;
unsigned long lastLoop=0;

// ─────────────────────────────────────────────────────
void setup() {
  Serial.begin(115200);
  Wire.begin();
  ina.begin();
  ds.begin();
  oled.begin(SSD1306_SWITCHCAPVCC, 0x3C);
  oled.clearDisplay(); oled.display();
  pinMode(PIN_GATE,   OUTPUT); digitalWrite(PIN_GATE,   LOW);
  pinMode(PIN_PWM_XL, OUTPUT); analogWrite(PIN_PWM_XL,  0);
  pinMode(PIN_BUTTON, INPUT_PULLUP);
  Serial.println(F("Solar BEMS starting..."));
}

// ─────────────────────────────────────────────────────
void loop() {
  if (millis() - lastLoop < LOOP_MS) return;
  lastLoop = millis();

  readSensors();
  evaluateSafety();
  if (!faultActive) {
    runMPPT();
    manageCharging();
    manageLoad();
    updateSoH();
  }
  updateOLED();
  logSerial();
}

// ─────────────────────────────────────────────────────
void readSensors() {
  vpv  = ina.getBusVoltage_V();
  ipv  = ina.getCurrent_mA() / 1000.0f;
  ppv  = vpv * ipv;
  int adc = analogRead(PIN_VBAT_ADC);
  vbat = (adc * VREF / 1023.0f) * VDIV_RATIO;
  ds.requestTemperatures();
  temp = ds.getTempCByIndex(0);
}

// ─────────────────────────────────────────────────────
void evaluateSafety() {
  if (temp > T_FAULT) {
    faultActive = true;
    analogWrite(PIN_PWM_XL, 0);
    digitalWrite(PIN_GATE, LOW);
    loadOn = false;
```

```
    state  = FAULT;
    return;
  }
  if (faultActive && temp <= T_RECOVER)
    faultActive = false;
}

// ─────────────────────────────────────────────────────
void runMPPT() {
  if (vpv < V_PV_MIN) { state = NIGHT; return; }
  float dp = ppv - ppvPrev;
  if (dp >= 0)  dutyCycle += MPPT_STEP;
  else          dutyCycle -= MPPT_STEP;
  dutyCycle = constrain(dutyCycle, MPPT_MIN, MPPT_MAX);
  analogWrite(PIN_PWM_XL, map(dutyCycle, 0,100, 0,255));
  ppvPrev = ppv;
}

// ─────────────────────────────────────────────────────
void manageCharging() {
  if (vpv < V_PV_MIN) return;  // night, skip
  if (vbat >= V_FULL) {
    analogWrite(PIN_PWM_XL, 0);
    state = FLOAT_FULL; return;
  }
  chargeMode = (vbat >= V_CC_CV); // false=CC  true=CV
  state = chargeMode ? MPPT_CV : MPPT_CC;
  // Coulomb counting — charge phase
  coulombs_mAh += ipv * (LOOP_MS / 3600000.0f) * 1000.0f;
}

// ─────────────────────────────────────────────────────
void manageLoad() {
  if (vbat < V_LVD_OFF && loadOn) {
    digitalWrite(PIN_GATE, LOW);
    loadOn = false; state = LOAD_OFF;
  } else if (vbat > V_LVD_ON && !loadOn) {
    digitalWrite(PIN_GATE, HIGH);
    loadOn = true;  state = LOAD_ON;
  }
}

// ─────────────────────────────────────────────────────
void updateSoH() {
  float C_NOMINAL_MAH = 6000.0f; // 3×2000mAh cells
  soc = constrain((vbat-9.0f)/(12.6f-9.0f)*100.0f, 0, 100);
  // SoH from measured vs nominal (simplified)
  if (coulombs_mAh > 0)
    soh = constrain((coulombs_mAh / C_NOMINAL_MAH)*100.0f, 0,100);
}

// ─────────────────────────────────────────────────────
const char* stateLabel() {
  switch(state){
    case FAULT:      return "FAULT!";
    case NIGHT:      return "NIGHT";
    case MPPT_CC:    return "CHG-CC";
    case MPPT_CV:    return "CHG-CV";
    case FLOAT_FULL: return "FULL";
    case LOAD_OFF:   return "LVD OFF";
    case LOAD_ON:    return "LOAD ON";
    default:         return "--";
  }
}
```

```
// ──────────────────────────────────────────────────────
void updateOLED() {
  oled.clearDisplay();
  oled.setTextColor(SSD1306_WHITE);
  oled.setTextSize(1);
  char buf[22];
  // Line 1: PV
  oled.setCursor(0,0);
  snprintf(buf,22,"PV %4.1fV %4.2fA %5.1fW",vpv,ipv,ppv);
  oled.print(buf);
  // Line 2: Battery
  oled.setCursor(0,10);
  snprintf(buf,22,"BAT %5.2fV  SoC %3.0f%%",vbat,soc);
  oled.print(buf);
  // Line 3: Temperature + SoH
  oled.setCursor(0,20);
  snprintf(buf,22,"T%4.1fC  SoH %5.1f%%",temp,soh);
  oled.print(buf);
  // Line 4: Status
  oled.setCursor(0,30);
  snprintf(buf,22,"Mode: %-10s",stateLabel());
  oled.print(buf);
  // Line 5: Duty cycle
  oled.setCursor(0,40);
  snprintf(buf,22,"MPPT duty: %3d%%",dutyCycle);
  oled.print(buf);
  oled.display();
}

// ──────────────────────────────────────────────────────
void logSerial() {
  Serial.print(F("Vpv:")); Serial.print(vpv,2);
  Serial.print(F(" Ipv:")); Serial.print(ipv,3);
  Serial.print(F(" Ppv:")); Serial.print(ppv,2);
  Serial.print(F(" Vbat:")); Serial.print(vbat,2);
  Serial.print(F(" T:")); Serial.print(temp,1);
  Serial.print(F(" SoC:")); Serial.print(soc,1);
  Serial.print(F(" SoH:")); Serial.print(soh,1);
  Serial.print(F(" Mode:")); Serial.println(stateLabel());
}
// ── End of firmware ──────────────────────────────────
```

APPENDIX D – BILL OF MATERIALS

The following table presents the complete Bill of Materials for the Solar BEMS implementation phase. Reference designators correspond to the system schematic in Appendix A. Quantities represent the components used in the assembled system.

Table D.1 Solar BEMS complete bill of materials — implementation phase

Ref	Description	Value / Spec	Qty	Package
ARD1	Arduino Mega 2560	ATmega2560, 16MHz, 5V	1	Dev board
U1	INA219 Power Monitor	I2C, 26V max, 0x40	1	Breakout module
U2	XL4015 CC/CV Buck	8–36V in, 0–5A adj.	1	Module (3-screw)
U4	LM2596 Buck Supply	4.5–50V in, 5V out	1	Module
Q1	IRLZ44N N-MOSFET	55V, 47A, Rds=22mΩ	1	TO-220
U3	DS18B20 Temp Sensor	−55 to +125°C, 1-Wire	1	Waterproof probe
OLED1	SSD1306 OLED Display	0.96″, 128×64px, I2C 0x3C	1	Module
BT1–BT3	18650 Li-ion Cells	3.7V nom, ≥2000mAh	3	Cylindrical
BH1	3S 18650 Holder	Series config, 3-cell	1	Plastic holder
BMS1	XR-121 3S 20A BMS	4.2V/cell OVP, 20A	1	PCB module
PV1	25Wp Solar Panel	Voc=24.62V, Vmp=20.84V	1	420×355mm
LOAD1	12V DC Fan	12V, ≤0.5A, brushless	1	80mm PC fan
F1	5A Fuse + Holder	250V, fast-blow	1	5×20mm glass
D1	1N5822 Schottky	40V, 3A, Vf≈0.32V	1	DO-201AD
D4	1N4007 Rectifier	1000V, 1A	1	DO-41
TVS1	P6KE24A TVS Diode	24V clamp, 600W peak	1	DO-204AC
C1	470µF / 35V Electrolytic	Low ESR, 105°C	1	Radial
C8	470µF / 25V Electrolytic	Battery bus bulk	1	Radial
C15	100µF / 25V Electrolytic	Load output cap	1	Radial
CMCU	100µF / 16V Electrolytic	MCU supply decoupling	1	Radial
C14	100nF / 50V Ceramic	ADC RC filter	1	0805 / Radial

R1,R2	4.7kΩ Resistors	I2C SDA/SCL pull-ups	2	1/4W axial
R5	4.7kΩ Resistor	DS18B20 DQ pull-up	1	1/4W axial
R8	10kΩ Resistor	Voltage divider top	1	1/4W axial
R9	3.3kΩ Resistor	Voltage divider bottom	1	1/4W axial
R10	100Ω Resistor	MOSFET gate series	1	1/4W axial
R11	10kΩ Resistor	MOSFET gate pull-down	1	1/4W axial
RLED	1kΩ Resistor	LED current limiter	1	1/4W axial
LED1	Green LED 3mm	Power-on indicator	1	3mm THT
J1,J3	Screw Terminals 2-pin	PV in / load out	2	5.08mm pitch

APPENDIX E – PERFBOARD LAYOUT MAP

The following tables describe the hole-by-hole component placement and wiring for all three perfboard assemblies. Coordinates use the notation [band][row][column], where band 1 = top third, band 2 = middle third, band 3 = bottom strip for Board 1. Board 2 and Board 3 use single-letter rows (A–X) and numeric columns.

Board 1 (9×15 cm) — Four bus rails must be laid before any components:

- GND spine: column 2, rows 1A to 3B — bare tinned wire.
- +V_PV rail: row 1E, columns 25 to 31.
- V_BATT spine: column 28, rows 2E to 2S.
- +5V rail: row 3A, columns 4 to 20.

Refer to the detailed hole-by-hole connection tables provided in the build guide documentation accompanying this thesis (available in the project GitHub repository). Each table entry specifies: component reference, pin name, hole coordinate, and the connection to be made on the solder side.

Appendix F — Hardware Assembly Photographs

The following photographs document the complete hardware assembly of the Solar BEMS across all three perfboard boards and the fully integrated system. All boards were constructed using double-sided perfboard (2.54 mm pitch) with colour-coded 24 AWG signal wiring and bare tinned wire power bus rails. Modules are mounted on 2.54 mm pin header sockets to allow replacement without desoldering. Photographs were taken with the system powered and operational.

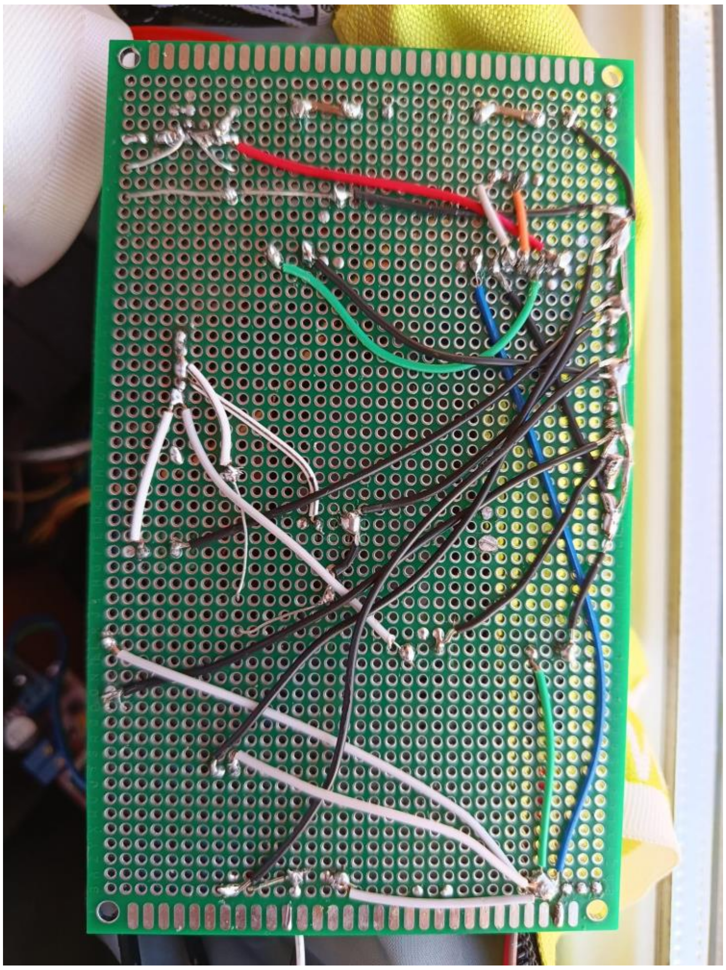

Figure F.1 Board 1 (9×15 cm) — power stage assembly

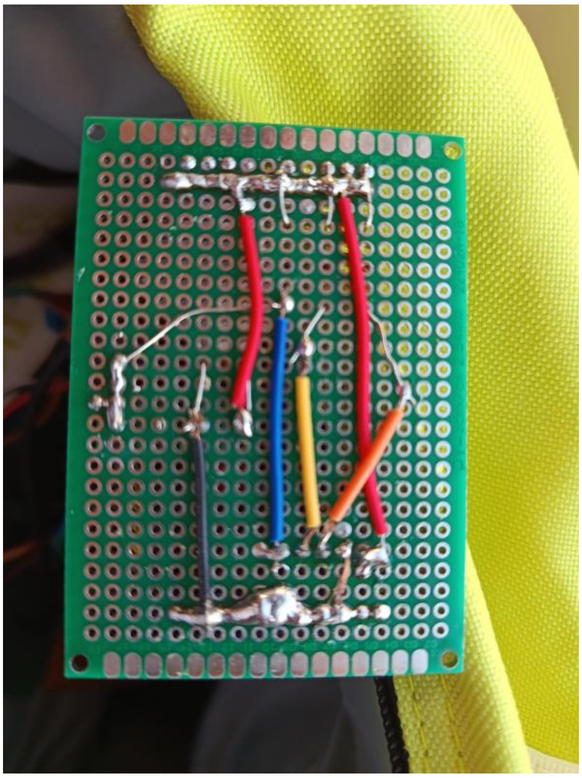
Figure F.2 Board 2 (5×7 cm) — sensor board

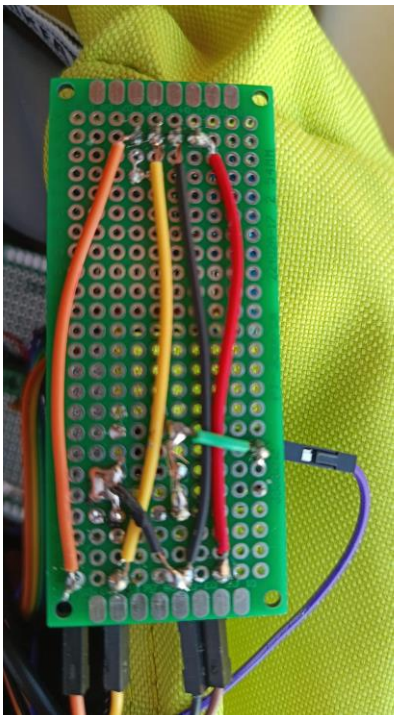
Figure F.3 Board 3 (3×7 cm) — display board

BIOGRAPHY

Name: Ahmed Salama

Date of Birth: 14/02/2003

Place of Birth: Saudi Arabia

Education:

2022 – 2026 Istanbul Arel University, Faculty of Engineering and Architecture, Electrical and Electronics Engineering (B.Sc.)

Projects:

- Solar Panel-Based Battery Energy Management System (Solar BEMS) — Graduation project. Designed and implemented a complete off-grid solar charging system with MPPT, CC/CV battery management, State-of-Health estimation, and real-time OLED monitoring on a three-board perfboard assembly. (2025–2026)
- Ball-Balancing Robot — STM32 microcontroller, PID control, second place out of fifteen teams in university robotics competition. Demonstrated real-time closed-loop control with encoder feedback and PWM motor drive. (2024–2025)
- Predictive Maintenance Simulation — Arduino with MPU6050 vibration sensor for bearing fault detection. Implemented FFT-based frequency domain analysis for fault signature identification. (2024)

Languages:

Arabic (native), English (B2 proficiency), Turkish (basic)

Internships: For my internship, I designed and implemented a Smart Parking Management System built around an Arduino Uno microcontroller. The work covered the full engineering cycle — component selection, breadboard prototyping, embedded firmware development in C++, and PCB design in KiCad.

The system uses two FC-51 infrared sensors to detect vehicles at a car park entrance and exit, an SG90 servo motor to control a physical gate, and a 16×2 I²C LCD, traffic-light LED module, and buzzer to communicate live parking status to drivers. The firmware is structured as a finite state machine with debounce timing and occupancy tracking to ensure reliable real-world operation.

The most significant technical challenge I encountered was a power-integrity fault: the servo motor's inrush current was causing voltage dips on the shared 5V rail, producing LCD glitches and occasional microcontroller resets. I diagnosed the issue through controlled testing and resolved it by incorporating a 470 µF bulk capacitor into a custom two-layer Arduino carrier shield designed in KiCad — converting a breadboard problem into a permanent hardware fix. The final system achieved 100% detection accuracy at the entrance sensor and 95% at the exit, with zero power-related faults following the hardware modification.